Week 5 PDF

CTRL A, CTRL C and CTRL V

AR Facial Recognition Glasses

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Introduction

This week we worked on Sprint 0 of our project on our facial recognition app for smart glasses. We worked on our Release Plan, Burndown Chart, Product Roadmap, did an overview of our System Architecture, justified our planning decisions and created our wireframes for the application.

Release Plan

AR App — Sprint Release Plan							
Sprint	Start Date	End Date	Duration	Story Points	Cumulative Done	% Complete	S
Sprint 0	03-Mar-26	10-Mar-26	7 days	5	5	3%	
Sprint 1	10-Mar-26	17-Mar-26	7 days	32	37	25%	P
Sprint 2	17-Mar-26	24-Mar-26	7 days	31	68	47%	P
Sprint 3	24-Mar-26	31-Mar-26	7 days	30	98	67%	P
Sprint 4	31-Mar-26	07-Apr-26	7 days	27	125	86%	P
Sprint 5	07-Apr-26	14-Apr-26	7 days	21	146	100%	P

a

Sprint Story Breakdown					
Key	Summary	Sprint	Points	Priority	Status
SCRUM-45	Monitor glasses battery & system status	Sprint 0	5	Medium	To Do
SCRUM-16	Facial recognition of known people	Sprint 1	8	Highest	To Do
SCRUM-6	Real-time display of name & relationship	Sprint 1	5	High	To Do
SCRUM-7	Face capture with explicit consent	Sprint 1	8	Highest	To Do
SCRUM-18	Relationship categorisation	Sprint 1	3	Medium	To Do
SCRUM-19	Unknown person detection & alert	Sprint 1	3	Medium	To Do
SCRUM-36	Secure storage of conversation data	Sprint 1	5	Highest	To Do
SCRUM-17	Automatic recognition of scanned faces	Sprint 2	10	High	To Do
SCRUM-32	Conversation recall on face recognition	Sprint 2	5	High	To Do
SCRUM-30	Automatic conversation transcription	Sprint 2	8	High	To Do
SCRUM-31	AI-generated conversation summaries	Sprint 2	8	High	To Do
SCRUM-20	Identification on demand (voice command)	Sprint 3	5	Medium	To Do
SCRUM-21	Add new contacts with permission	Sprint 3	5	High	To Do
SCRUM-22	Object identification (keys, phone, medication)	Sprint 3	10	Lowest	To Do
SCRUM-23	Voice recognition fallback (low light)	Sprint 3	7	Low	To Do
SCRUM-34	Conversation prompts (short, optional)	Sprint 3	3	Medium	To Do
SCRUM-43	Emergency call feature	Sprint 4	8	Low	To Do
SCRUM-47	All stored data encrypted	Sprint 4	8	Highest	To Do
SCRUM-49	Secure personal information handling	Sprint 4	8	High	To Do

SCRUM-54	Comfortable everyday display	Sprint 4	3	High	To Do
SCRUM-33	Memory journal (photos, videos, captions)	Sprint 5	8	Medium	To Do
SCRUM-46	Biometric data consent & withdrawal	Sprint 5	5	Highest	To Do
SCRUM-52	Automatic system, minimal interaction	Sprint 5	8	High	To Do
SCRUM-55	Directional guidance overlaid on vision	Backlog	TBD	Lowest	To Do
SCRUM-56	View to-do list on glasses	Backlog	TBD	Low	To Do
SCRUM-57	Show date and time on standby	Backlog	TBD	High	To Do
SCRUM-58	Caregiver monitoring via glasses	Backlog	TBD	Low	To Do
SCRUM-59	Family manage memory features via external app	Backlog	TBD	Low	To Do
SCRUM-60	Family disable memory prompts	Backlog	TBD	Medium	To Do
SCRUM-61	Caregiver manage contact database remotely	Backlog	TBD	High	To Do
SCRUM-62	Family/carer manage settings & approve data	Backlog	TBD	Medium	To Do
SCRUM-48	Stress/upset detection & alert trusted person	Backlog	8	Low	To Do
SCRUM-50	Calming content when anxious	Backlog	7	Lowest	To Do
SCRUM-51	Conversation starters during social situations	Backlog	5	Low	To Do
SCRUM-35	Conversation starters during pauses	Backlog	5	Medium	To Do
SCRUM-53	Voice recognition of speakers	Backlog	TBD	Low	To Do
SCRUM-8	Store person name, relationship, notes	Backlog	TBD	High	To Do
SCRUM-14	Step-by-step reminders for daily tasks	Backlog	TBD	Low	To Do
SCRUM-44	remove later (placeholder)	Backlog	TBD	Medium	Done

Burndown Chart

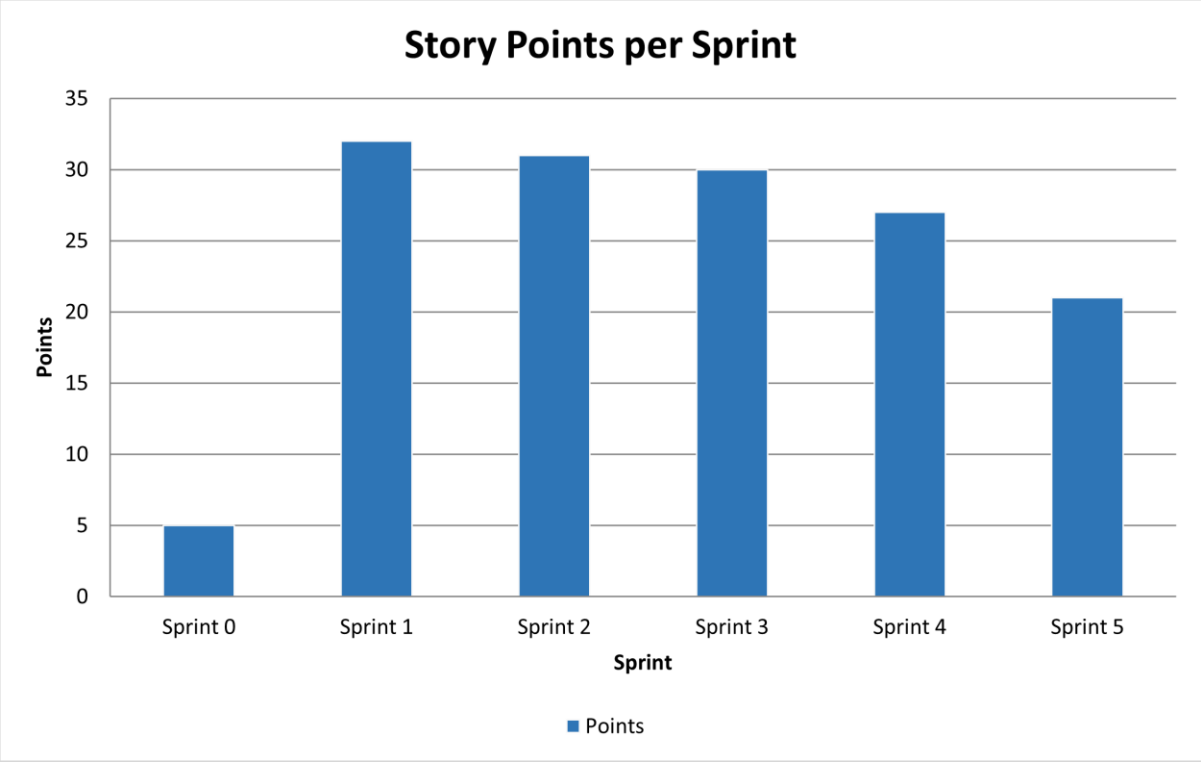
AR App — Sprint Burndown Chart

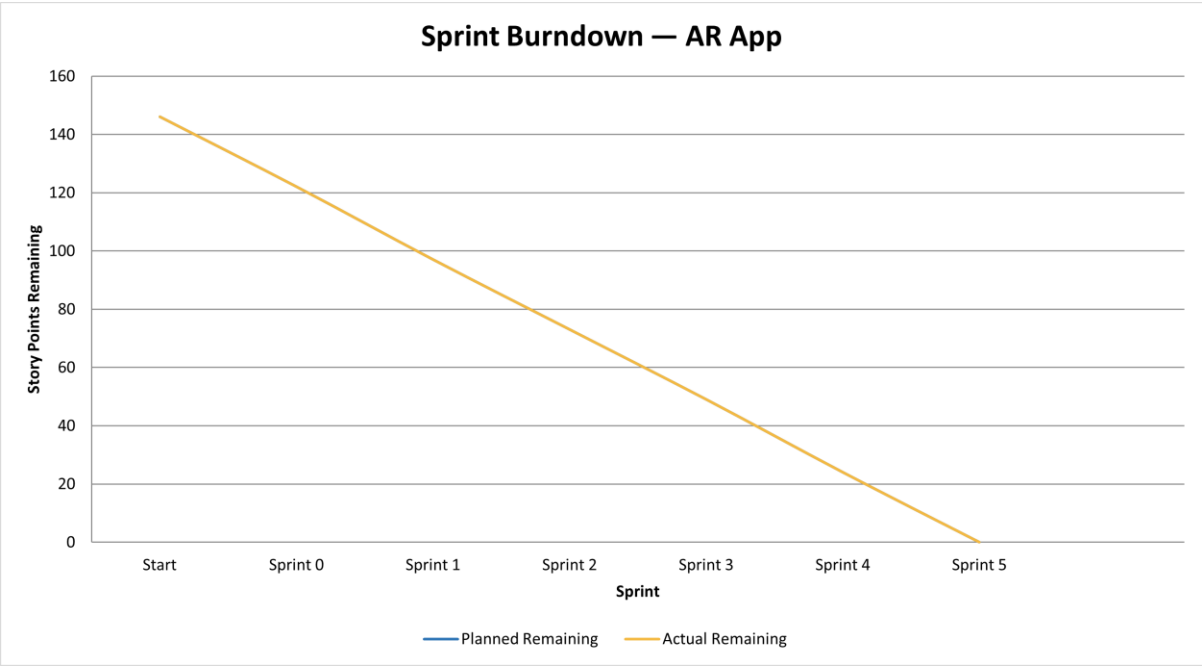
Burndown Data

		Planned Remaining	Actual Remaining	Points Completed
Start	02-Mar-26	146	146	0
Sprint 0	03-Mar-26	122	122	5
Sprint 1	10-Mar-26	97	97	32
Sprint 2	17-Mar-26	73	73	31
Sprint 3	24-Mar-26	49	49	30
Sprint 4	31-Mar-26	24	24	27
Sprint 5	07-Apr-26	0	0	21

Sprint Points Summary	
Sprint	Points

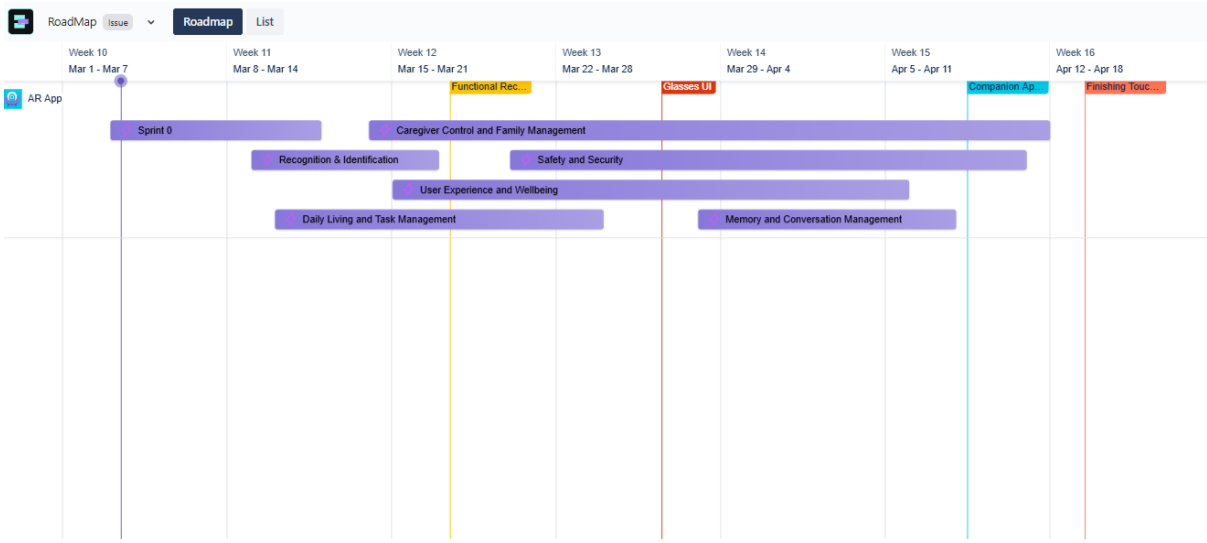
Sprint 0	5
Sprint 1	32
Sprint 2	31
Sprint 3	30
Sprint 4	27
Sprint 5	21
TOTAL	**146**





Product Roadmap

We started Sprint 0 mid way through Week 10 and will be finishing this week where we will be setting up our IDE's and resources we will use for our project and build a picture of what exactly we want from our project.



In Week 11 we will start working on our Recognition and Identification and Daily Living and Task Management epics. This will be the start of our work on functional facial

recognition for the application which we will work on until mid-way through Week 12. 2 people will work on each epic.

Week 12 will mark the end of our work on the Recognition and Identification Epic and thus the Functional Facial Recognition will conclude and be the start of our work on Caregiver Control and Family Management epic, User Experience and Wellbeing epic and we will start our Safety and Security epic at the end of the week. We will each work on an individual epic to spread the workload evenly. This marks the start of our work on the glasses UI.

In Week 13 our work on the Daily Living and Task Management epic and our Glasses UI will conclude and we start working on our Memory and Conversation Management epic. This will be the start of our work on the mobile companion app that will accompany our application for the smart glasses.

Week 15 will conclude the work on all our epics and the companion app so we can use the time in Week 16 for our finishing touches and adjustments for the app.

System Architecture Overview

React Native App Layer (Shared)

This is the core of the Joggo App, and it runs on both iOS and Android. It will handle:

• UI/UX: Screens for reminders, face identification results, mobile app etc.

• Navigation: React Navigation to move between screens.

• Firebase: Will securely store all our user's data in a database.

• Native Module Bridge: Calls our code for the AR glasses using Reacts Native's Module API.

Android AR Layer (Java & C++)

ARCore is Google's AR Framework and supports both java and C++ natively.

• Java (ARCore SDK): Manages the AR session lifecycle, anchors, plane detection, and augmented image tracking.

• C++ (ARCore C API): Handles tasks that are critical like camera frame processing and real time object/face tracking.

• JNI Bridge: This will connect Java and C++ layers so they can pass AR data between each other.

• React Native: Shows the user the AR results for example recognised face this is John your son.

IOS AR Layer (Swift & ARKit)

ARKit is apples framework and integrates tightly with Swift.

• ARSCNView / AR Session: Manages the AR camera session and scene Rendering.

• ARImageTrackingConfiguration: For tracking photos of people, they should remember.

• Swift Native Module: Wraps ARKit results and allows React Native to show the user the information the glasses are receiving.

Firebase

Service	What it will do.
Firestore	Store profiles, reminders, recognised person data
Firebase Auth	Caregiver/User login and role management.
Firebase Storage	Store reference photos of faces/ objects for AR tracking.
Could Functions	Trigger Notifications (For Example Missed Medication)

Justification for Planning.

Sprint Length

As a team we have decided to keep our sprints to one week. We feel like this will be enough time to complete the sprints that we have planned out so far. At the end of each sprint, we will hold a short review to demo what we did and discuss what went well and what we can improve on. This means our velocity is not fixed and will be adjusted each sprint if needed.

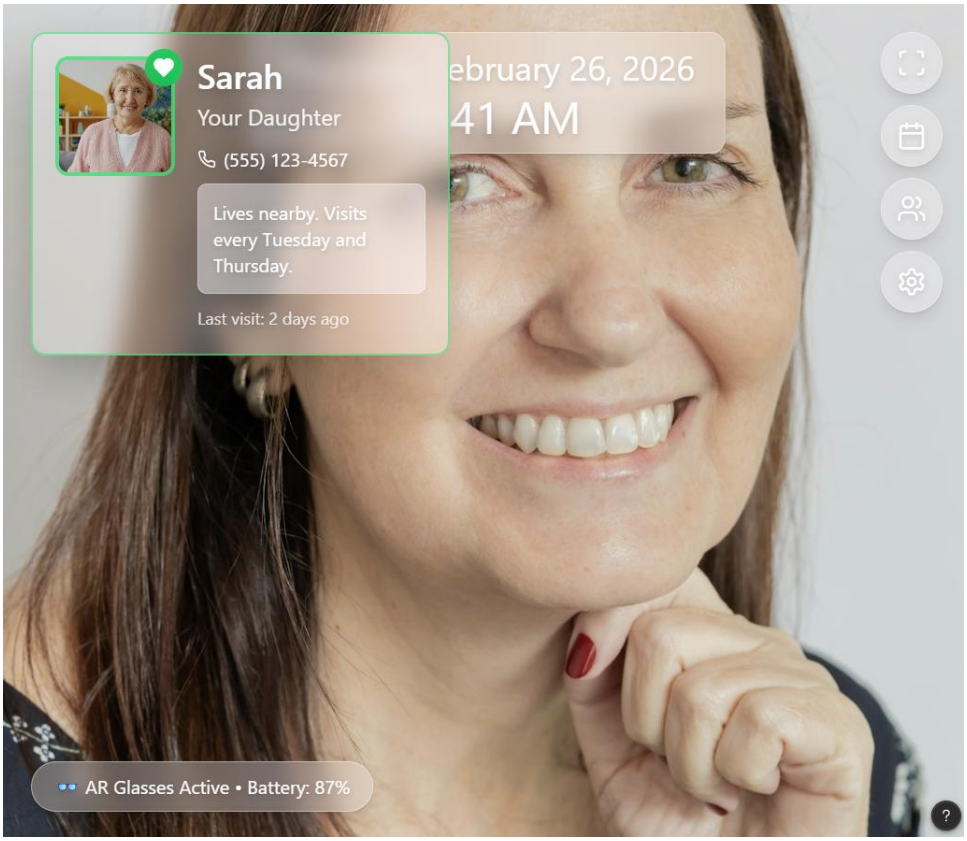
Team Velocity

We discussed in a group meeting that we could all get 8 story points done a week with this information we could multiply 8 points by the amount of people in the group and it is 32, our group velocity is 32 story points a week. We used planning poker to estimate each story point, considering the complexity and effort it would take.

Total Project Capacity

With our velocity of 32 story points a week, over the course of the project we will have a total capacity of 224 story points. This allows us to confirm the backlog is achievable within the timeline confirming our project is in scope.

Wireframes



Glasses GUI

Initial Mock Up

AI Enhanced Mock Up

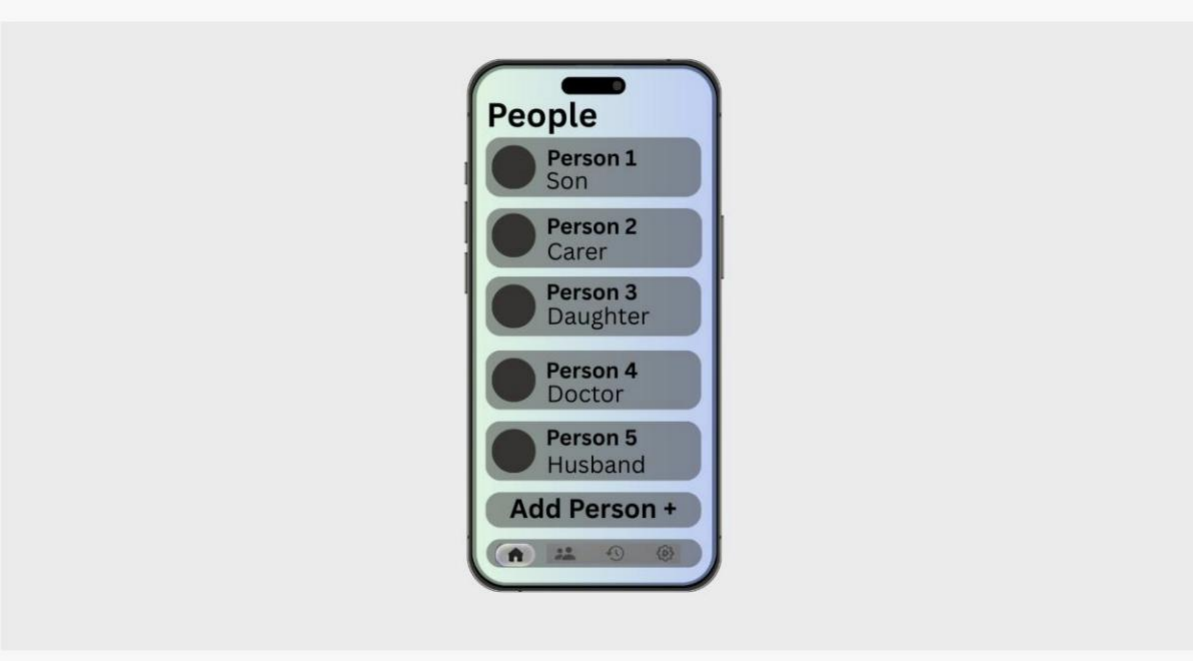
App Home Page

GUI Initial Mock Up

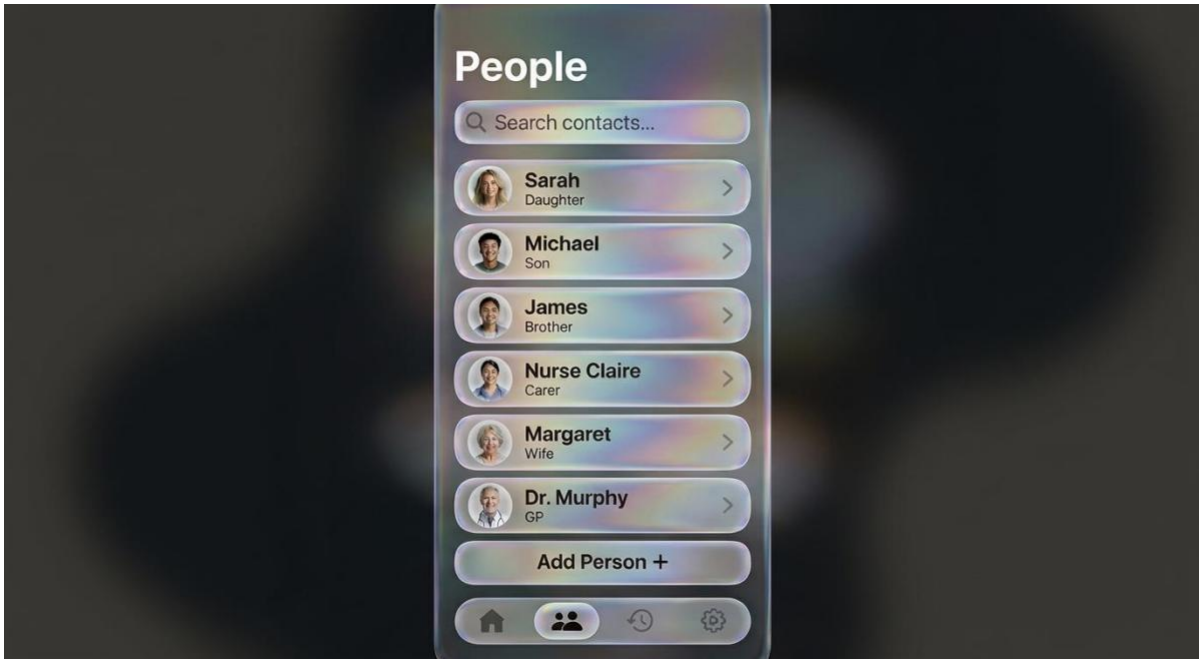
AI Enhanced Mock Up

App People Page

GUI Initial Mock Up



AI Enhanced Mock Up



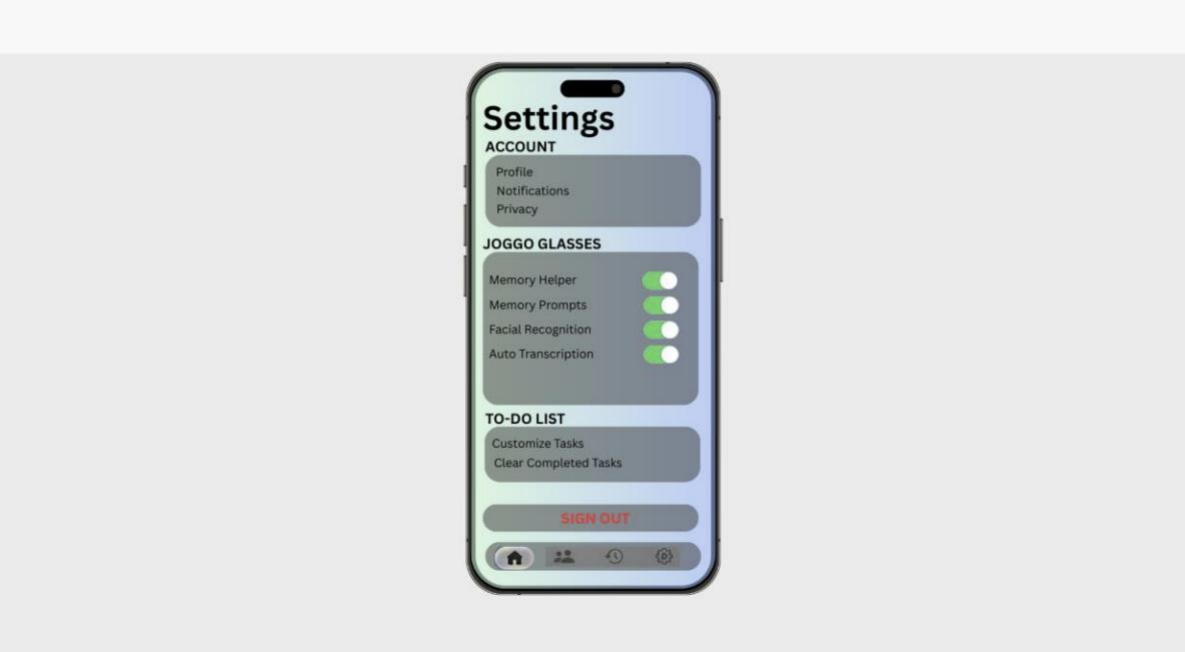
App History Page

GUI Initial Mock Up

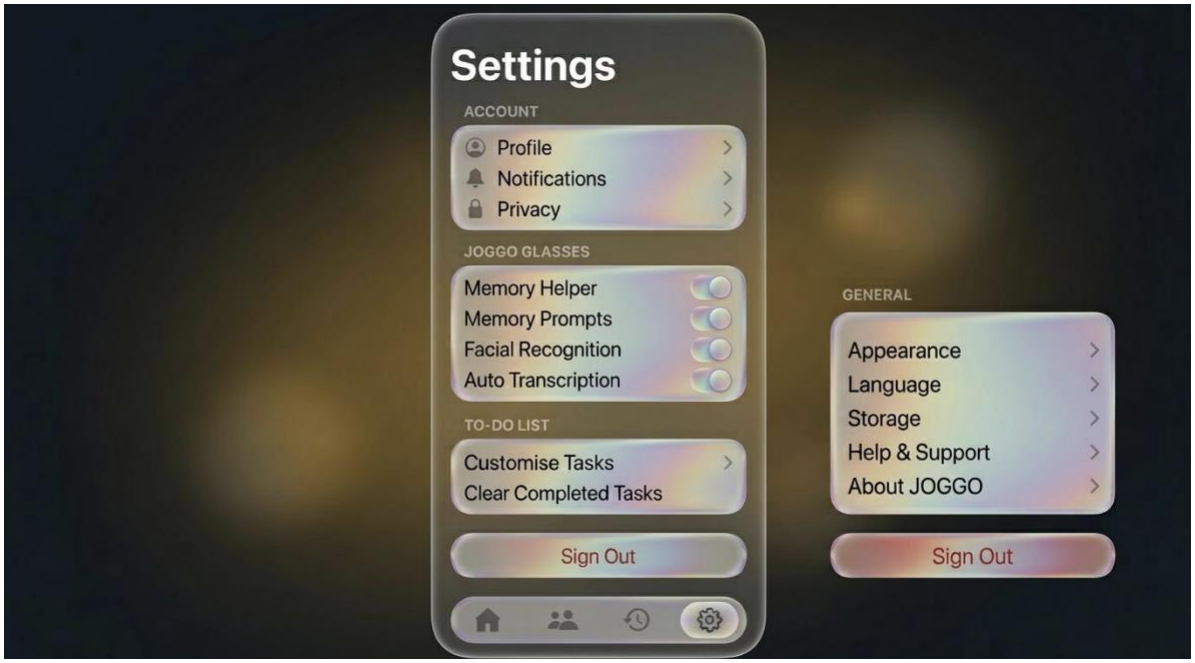
AI Enhanced Mock Up

App Settings Page

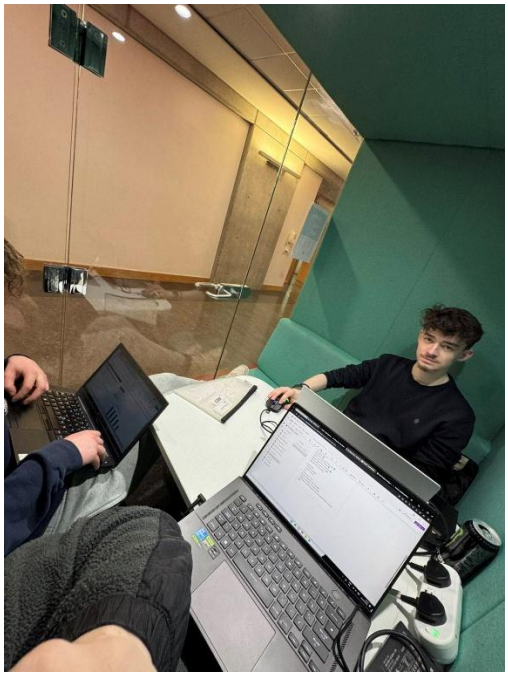
GUI Initial Mock Up

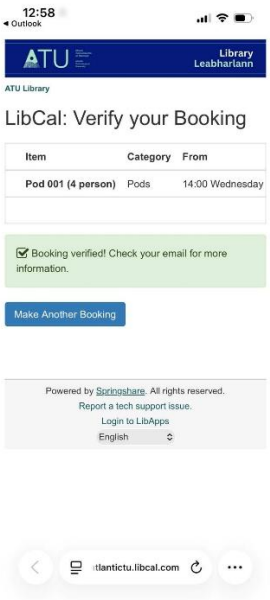


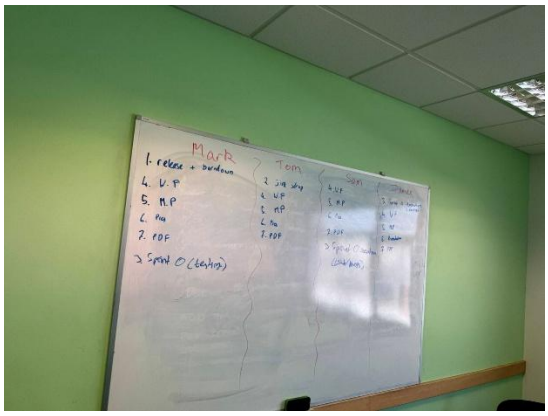
AI Enhanced Mock Up



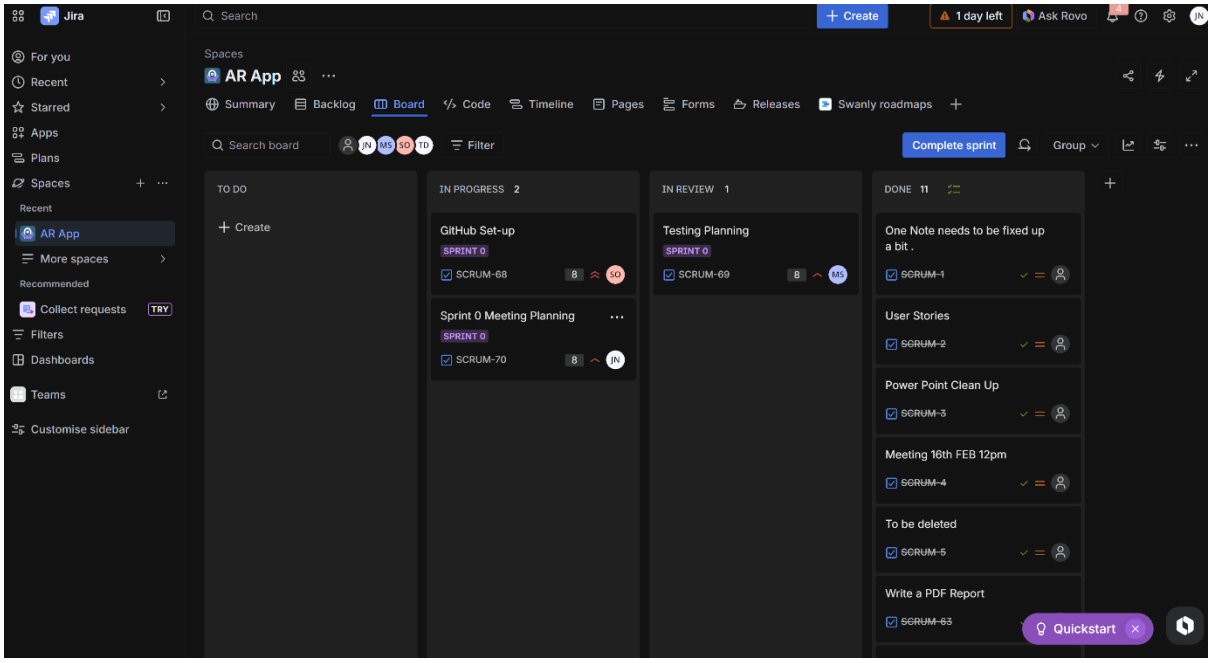
Evidence of Meetings

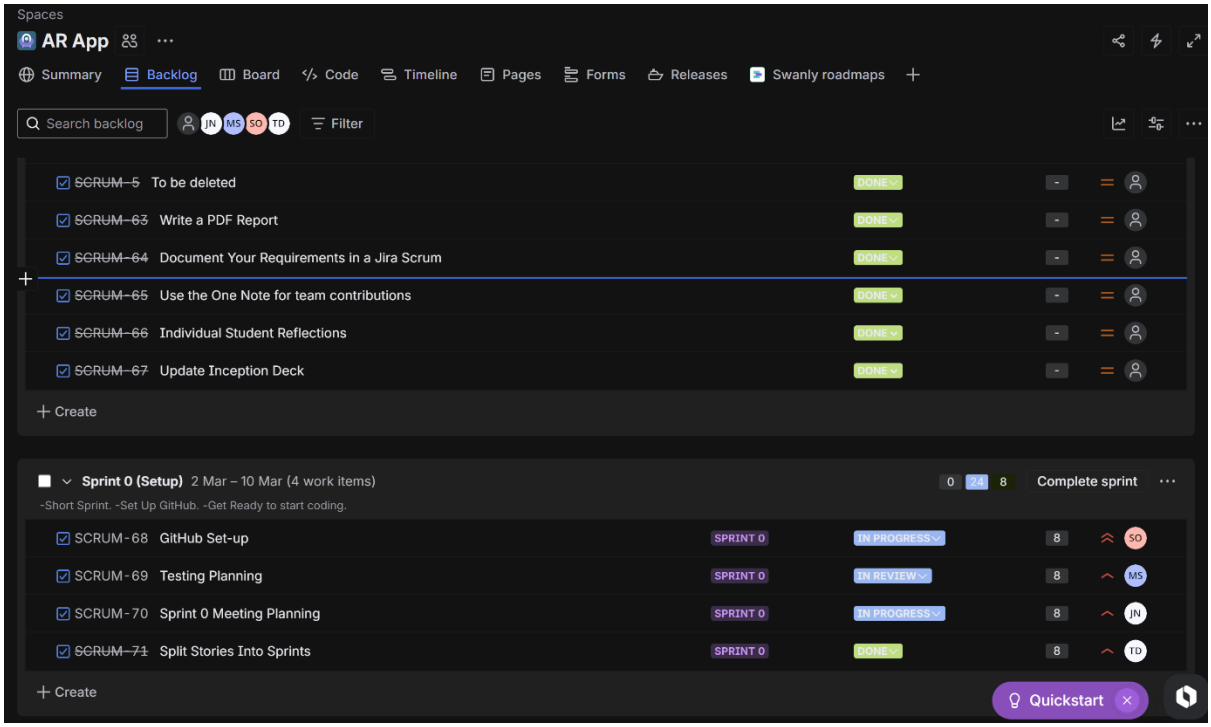


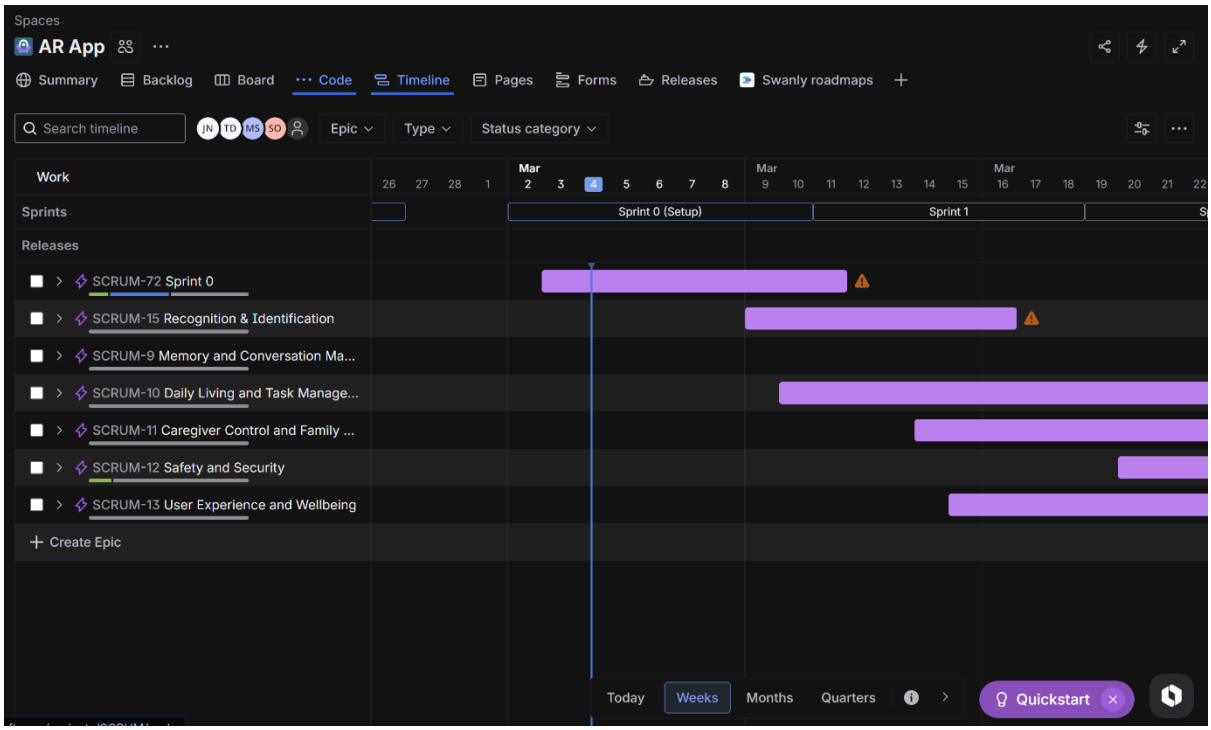




Jira and Gantt Chart




Spaces
AR App
Summary
Backlog
Board
Code
Timeline
Pages
Forms
Releases
Swanly roadmaps
Search backlog
Filter
SCRUM-5 To be deleted
SCRUM-63 Write a PDF Report
SCRUM-64 Document Your Requirements in a Jira Scrum
SCRUM-65 Use the One Note for team contributions
SCRUM-66 Individual Student Reflections
SCRUM-67 Update Inception Deck
DONE
Create
Sprint 0 (Setup) 2 Mar – 10 Mar (4 work items)
-Short Sprint. -Set Up GitHub. -Get Ready to start coding.
Complete sprint
SCRUM-68 GitHub Set-up
SCRUM-69 Testing Planning
SCRUM-70 Sprint 0 Meeting Planning
SCRUM-71 Split Stories Into Sprints
SPRINT 0
IN PROGRESS
IN REVIEW
Quickstart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Swanly roadmaps
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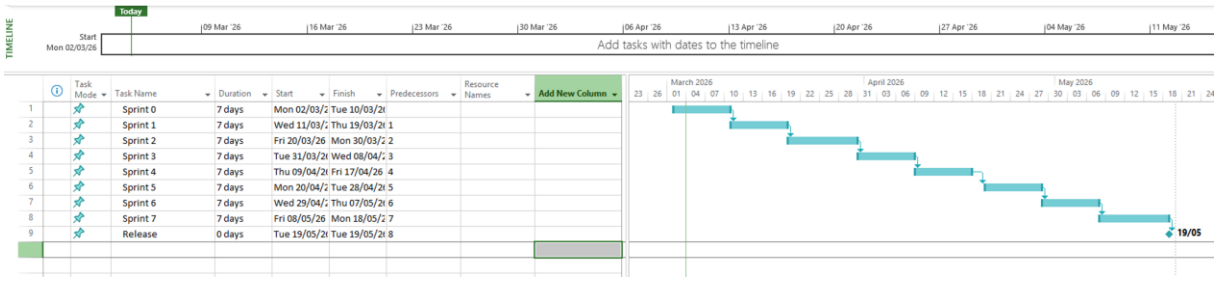
Epic
Type
Status category
Work
Sprints
Releases
Mar
Sprint 0 (Setup)
Sprint 1
SCRUM-72 Sprint 0
SCRUM-15 Recognition & Identification
SCRUM-9 Memory and Conversation Ma...
SCRUM-10 Daily Living and Task Manage...
SCRUM-11 Caregiver Control and Family ...
SCRUM-12 Safety and Security
SCRUM-13 User Experience and Wellbeing
Create Epic
Today
Weeks
Months
Quarters
Quickstart


TIMELINE
Today
Start
Mon 02/03/26
Add tasks with dates to the timeline
Task Mode
Task Name
Duration
Start
Finish
Predecessors
Resource Names
Add New Column
Sprint 0
Sprint 1
Sprint 2
Sprint 3
Sprint 4
Sprint 5
Sprint 6
Sprint 7
Release
March 2026
April 2026
May 2026
19/05

Links and References

Jira

GitHub

One Note

Microsoft Project File: https://atlantictu-my.sharepoint.com/personal/g00427970_atu_ie/Documents/ATU_Gal_23_24/Year%203/Sem%202/Project%20Management/GanttChart.png